

Track record analysis of an institutional private equity portfolio

Realised versus unrealised value and benchmarking against public markets · CalPERS Private Equity Program, as of 31 December 2025 · prepared 15 September 2026

Summary

The analysis covers 466 funds from CalPERS' published private equity portfolio, with 152 bn USD committed and 122 bn USD paid in. Two findings emerge, and they differ in how firmly they can be stated.

The robust finding concerns the composition of value, not its size. For vintages 2005-2013, 1.57x of paid-in capital has come back as cash against a total value of 1.66x. For vintages 2016-2020 the total value is almost unchanged at 1.62x, but only 0.64x has been realised. An unchanged TVPI therefore masks a shift from realised to unrealised value, and unrealised value is the manager's own mark, not a market price.

The uncertain finding is the excess return. Measured as a public market equivalent against the S&P 500 total return, the portfolio's mature vintages sit between 0.84 and 1.03 depending on which timing assumption is applied. The range crosses 1.00. Whether the portfolio as a whole beat public markets therefore cannot be settled on the publicly available data. For individual vintages it can: 10 of 15 sit below the index in all three scenarios, 2 sit above in all three, and 3 depend on the assumption.

The data

CalPERS publishes its private equity portfolio at fund level with fund name, vintage year, capital committed, paid-in capital, distributions, total value including remaining holdings, net IRR and multiple. The data carries one decisive limitation: **there are no dated cash flows**. Only totals as of the valuation date are public. That determines what can be computed faithfully, and it is why the benchmarking below is reported as a range rather than a single number.

Of the 466 funds, 326 carry a reported net IRR. The remainder are too young for a return figure to be meaningful and are marked N/M in the source.

Data validation

The table is extracted programmatically from the published CalPERS document, and two checks stand between the source and the analysis.

Row level. Every row is cross-checked against the source's own multiple column: computed TVPI, that is total value divided by paid-in capital, must match the reported multiple within rounding. Remaining value is checked as non-negative and paid-in capital as positive. All 466 rows pass.

Aggregate level. CalPERS states the portfolio's aggregate net multiple as 1.5x in the document's opening. The load produces 1.55x. That figure appears in no individual row, so the agreement confirms the extract as a whole rather than merely the rows' internal consistency.

Method

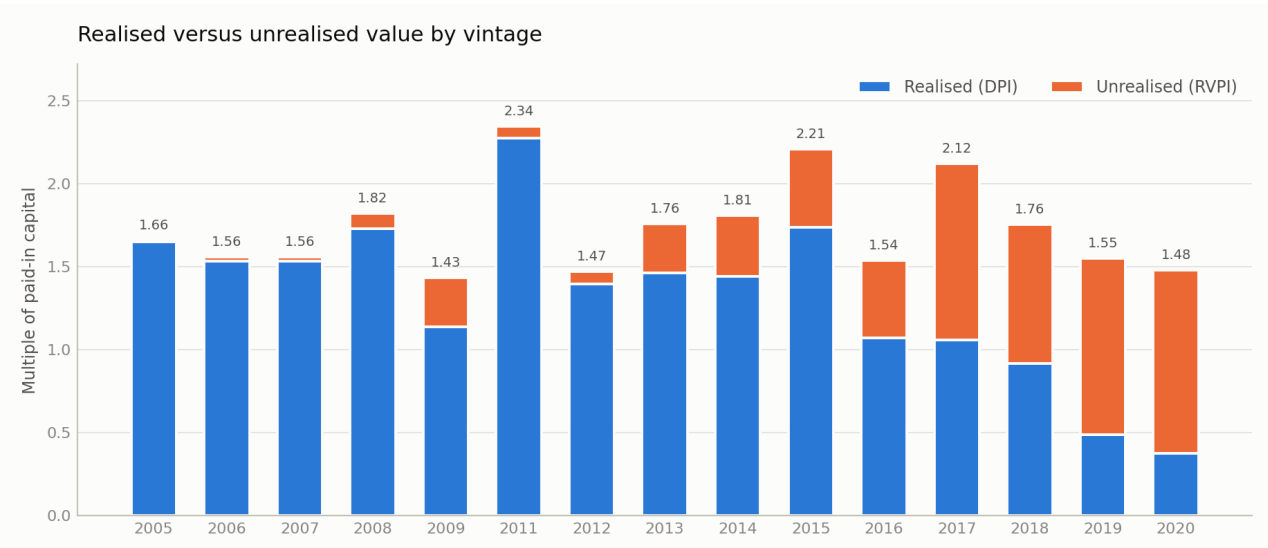
A public market equivalent compares a fund's cash flows with the same capital invested in a public index at the same times. All flows are carried forward to the valuation date at the index return, and the ratio of carried-forward distributions plus remaining value to carried-forward contributions is the PME. A value above 1.00 means outperformance against the index.

Because dated flows do not exist, the observed totals are distributed across time according to an assumed profile. **The totals are data; the distribution is an assumption.** PME is therefore computed under three profiles

— fast drawdown with early exits, a base case, and slow drawdown with late exits — and the result is reported as the range between them. If the whole range sits on one side of 1.00, the sign is robust to the assumption. If the range crosses 1.00, the conclusion does not hold, and that is stated rather than picking the scenario that suits best.

The benchmark is the S&P 500 total return. That is a demanding choice for vintages from 2009 onward, when US equities ran exceptionally, and it depresses PME for those years in particular. A broader or more comparable index would produce different levels. The benchmark choice is itself an assumption, on a par with the timing profile.

Realised versus unrealised value

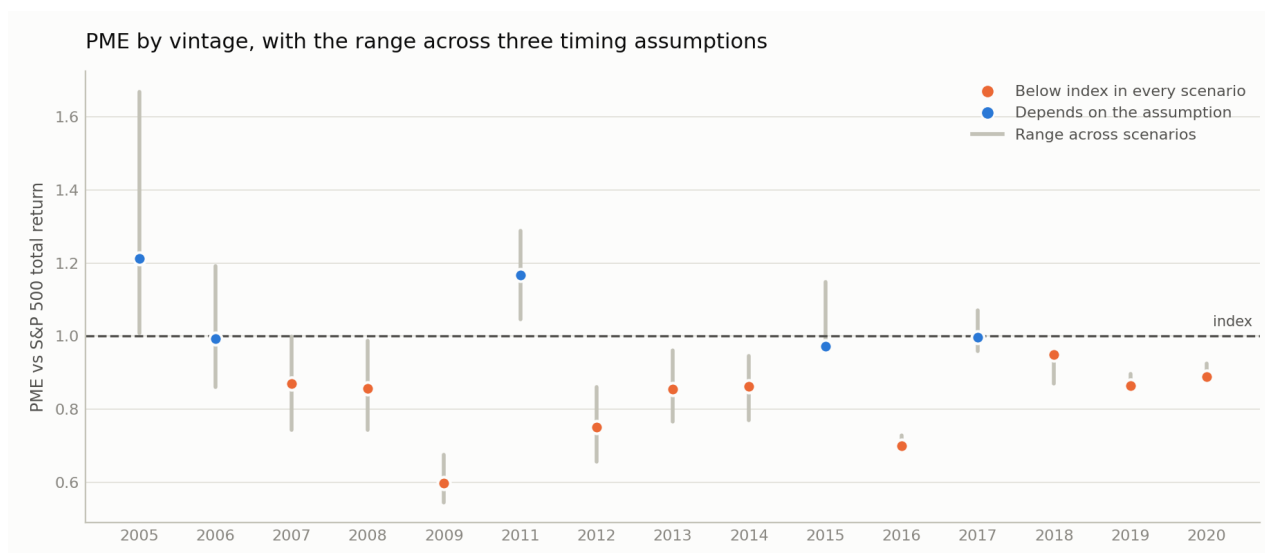


Bars show capital-weighted DPI and RVPI by vintage. The figure above each bar is TVPI. Only vintages through 2020 are shown; later years are too young for the split to be meaningful.

The pattern is unambiguous. Vintages from 2005 to 2008 are largely fully realised: the value has come back as cash. From 2016 onward, unrealised value is the majority, and for 2019 and 2020 less than half of paid-in capital has been distributed. Comparing TVPI across the two groups therefore does not compare like with like: for the early vintages the figure is an account, for the later ones it is a forecast.

For an investor with commitments across several fund generations, this is the difference between liquidity and paper value, and it is precisely the distinction a manager assessment based on TVPI alone will miss.

Benchmarking against public markets



The point is the base case; the vertical bar is the range across all three timing assumptions. Orange marks the vintages where the entire range sits below 1.00.

10 of 15 vintages shown sit below the index regardless of which timing assumption is used, including the vintages around the financial crisis and the middle of the 2010s. For those the conclusion is robust: the same capital in the S&P 500 would have returned more.

At the same time the range for several vintages is wide enough to cross 1.00, and for the oldest vintages it is very wide, because a small shift in assumed timing compounds over twenty years of index returns. This is not noise to be explained away — it is the real uncertainty that the absence of dated cash flows imposes on any PME computed from public data.

What the analysis cannot say

Whether the portfolio as a whole beat public markets. The aggregate PME range of 0.84 to 1.03 crosses 1.00.

Anything about risk. PME does not adjust for leverage, illiquidity or sector exposure. Two portfolios with the same PME can carry very different risk profiles.

Whether unrealised values are fair. Remaining value is the manager's own mark. For the most recent vintages it is the majority of reported value, and the analysis takes it at face value.

Anything about manager quality. The data carries no strategy, geography or sector, and a single LP's slice of a manager's funds is not that manager's track record.

Appendix: vintage summary

Vintage	Funds	Paid in (m USD)	TVPI	DPI	Unrealised share
2005	5	904	1.66x	1.65x	0 %
2006	12	5,093	1.56x	1.54x	2 %
2007	23	7,178	1.56x	1.53x	2 %
2008	13	4,501	1.82x	1.73x	5 %
2009	3	661	1.43x	1.14x	21 %
2011	6	1,286	2.34x	2.28x	3 %
2012	7	2,748	1.47x	1.40x	5 %
2013	9	3,801	1.76x	1.46x	17 %
2014	7	2,488	1.81x	1.44x	20 %
2015	12	2,294	2.21x	1.74x	21 %
2016	8	3,068	1.54x	1.07x	30 %
2017	9	3,186	2.12x	1.06x	50 %
2018	15	3,845	1.76x	0.92x	48 %
2019	22	8,027	1.55x	0.49x	68 %
2020	22	9,482	1.48x	0.37x	75 %

Unrealised share is RVPI divided by TVPI, that is how much of the reported value has not yet been realised.

Sources and reproducibility

Fund data: CalPERS Private Equity Program Fund Performance Review, as of 31 December 2025. Index data: S&P 500 annual total returns. The method follows Kaplan and Schoar (2005) for PME, adapted as described under Method. All data processing is in Python with SQLite as the data layer; the calculations can be reproduced from the source files.